



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO LAKSHMI AGENCIYES

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

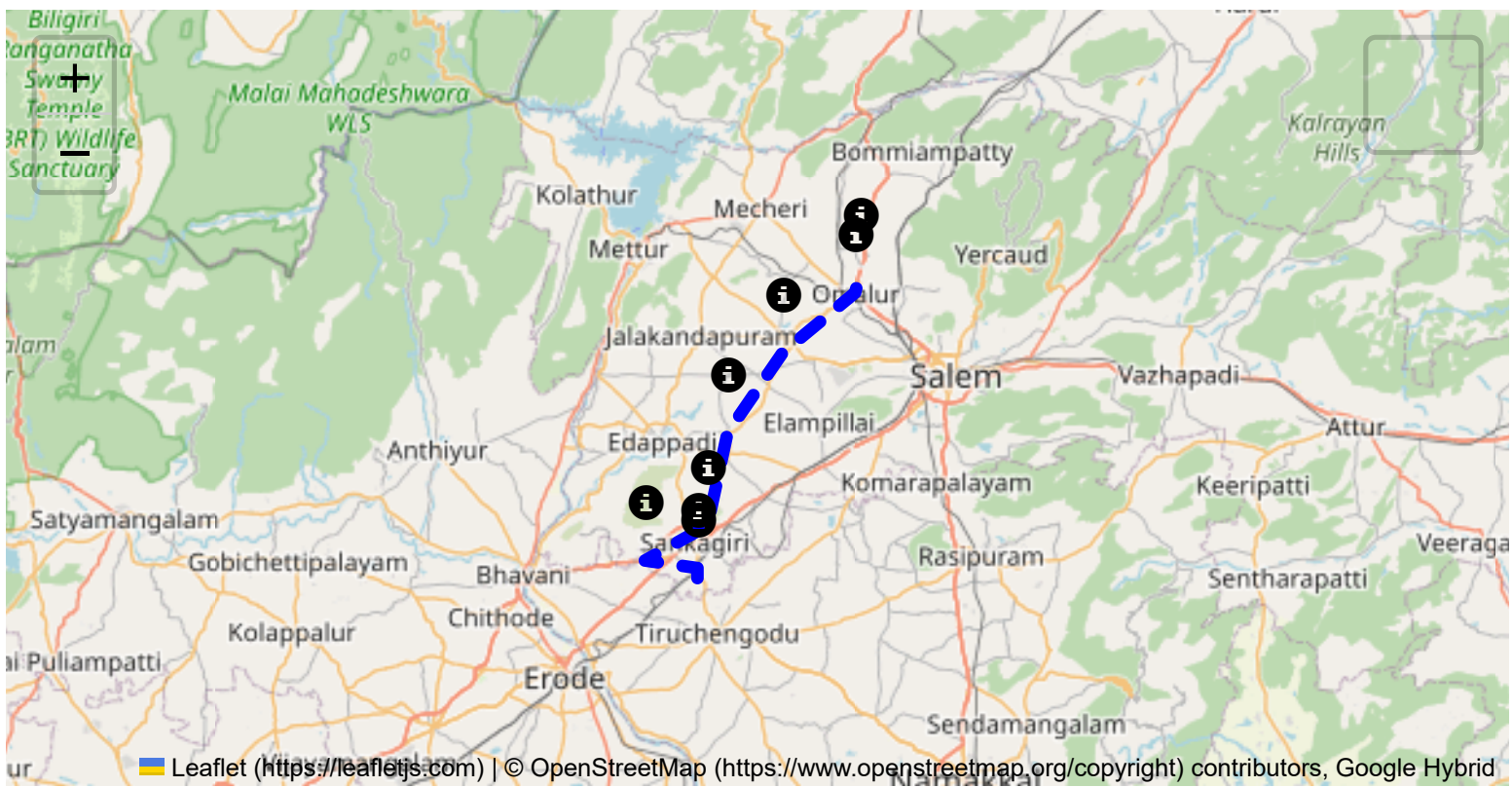
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

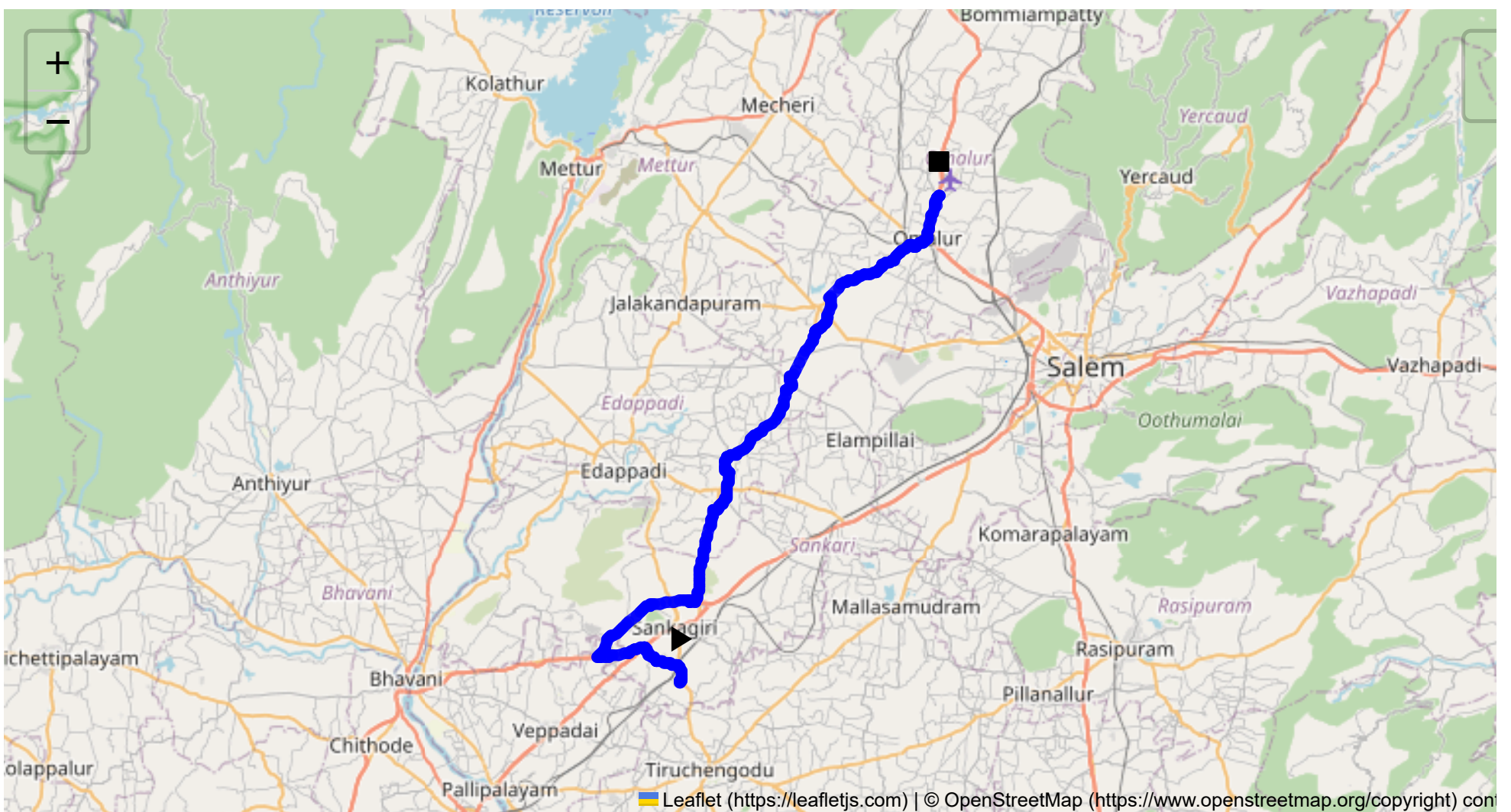
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 58.58 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.765436, 78.053238)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri, Tamil Nadu to Kamalapuram passes through various towns and landmarks, including Padaiveedu, Edappadi, Desavilakku, and Omalur. The journey is approximately 58.58 kilometers long, primarily following local roads and highways.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with moderate to heavy monsoon rains, especially between June and September. The route may face rainfall-induced hazards such as waterlogging, slippery roads, and reduced visibility. During summer (March to June), high temperatures may affect road conditions and vehicle performance.

3. Traffic Patterns, Peak Hours, and Congestion

Traffic in these areas typically peaks during morning hours (7-9 AM) and late afternoon to evening (5-8 PM), with congestion higher in urban areas like Sangagiri and Omalur. Expect slower movement on narrow roads, especially near marketplaces and school zones.

4. Assessment of Road Quality and Infrastructure

The route is a mix of local roads and highways, with variable quality. Some stretches may have potholes and uneven surfaces, particularly in rural areas. Urban sectors have better infrastructure but are prone to congestion.

5. Suggestions for Alternative Routes in Emergencies

In emergencies, consider rerouting via NH 544, which provides a slightly longer but more robust infrastructure, or utilize main roads that parallel the original route, offering better emergency response facilities.

6. Summary of Local Regulations Affecting Hazardous Material Transport

In Tamil Nadu, transporting hazardous materials requires adherence to the Motor Vehicles Act and Central Motor Vehicle Rules. Specific permits and safety measures must be observed. Checkpoints for compliance are more frequent on highways.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

There have been occasional accidents involving heavy vehicles due to overloading and brake failures. However, incidents involving hazardous materials have been rare, thanks to improved regulatory oversight.

8. Environmental Considerations and Sensitive Areas

The route traverses near agricultural fields and water bodies, which are sensitive to pollution from hazardous materials. Transfers and stops should be carefully managed to prevent spills or contamination.

9. Communication Coverage and Potential Dead Zones

Mobile network coverage is generally good but can be patchy in remote rural areas, especially around Desavilakku. Equip vehicles with reliable GPS and communication devices.

10. Estimated Emergency Response Times

Emergency response is quicker in towns (within 30 minutes), but rural segments may experience longer wait times due to limited accessibility. Plan for sufficient emergency protocols and contacts along the route.

11. Overall Summary of Risk Assessment

- Weather Hazards:** Moderate risk during monsoon.
- Traffic and Congestion:** Moderate risk, with urban areas presenting slowdowns.
- Road Quality:** Variable, with potential hazards in rural stretches.
- Emergency Preparedness:** Requires improved communication in some rural sections.
- Regulatory Compliance:** Essential for safe and lawful transportation of hazardous materials.

Overall, while the route is manageable within a relatively short time frame, preparation for weather conditions and maintaining compliance with safety standards are critical. Regular monitoring of local traffic updates and contingency planning are advisable for a safe journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	Medium	11.43822, 77.87348	30 KM/Hr
1	Turn	High	11.43968, 77.87345	15 KM/Hr
2	Turn	High	11.44029, 77.87544	15 KM/Hr
3	Turn	Medium	11.44898, 77.87410	30 KM/Hr
4	Turn	Medium	11.45357, 77.85789	30 KM/Hr
5	Turn	Medium	11.45352, 77.85698	30 KM/Hr
6	Turn	Medium	11.45847, 77.85140	30 KM/Hr
7	Turn	Medium	11.46048, 77.85036	30 KM/Hr
8	Turn	Medium	11.46303, 77.84945	30 KM/Hr
9	Turn	Medium	11.46314, 77.84928	30 KM/Hr
10	Turn	High	11.45542, 77.81725	15 KM/Hr
11	Turn	High	11.45631, 77.81668	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.49409, 77.88004	15 KM/Hr
13	Turn	Medium	11.49419, 77.88015	30 KM/Hr
14	Turn	High	11.49383, 77.88442	15 KM/Hr
15	Turn	Medium	11.49419, 77.88454	30 KM/Hr
16	Turn	Medium	11.49448, 77.88490	30 KM/Hr
17	Turn	High	11.55608, 77.89762	15 KM/Hr
18	Turn	Medium	11.58294, 77.90775	30 KM/Hr
19	Turn	High	11.58320, 77.90646	15 KM/Hr
20	Turn	Medium	11.64197, 77.95197	30 KM/Hr
21	Turn	Medium	11.67858, 77.96834	30 KM/Hr
22	Turn	High	11.68921, 77.97737	15 KM/Hr
23	Turn	Medium	11.68927, 77.97749	30 KM/Hr
24	Turn	High	11.70223, 77.97860	15 KM/Hr
25	Turn	Medium	11.71918, 78.01127	30 KM/Hr
26	Turn	Medium	11.72565, 78.01770	30 KM/Hr
27	Turn	Medium	11.73651, 78.03873	30 KM/Hr
28	Turn	Medium	11.73658, 78.03897	30 KM/Hr
29	Turn	Medium	11.74271, 78.04717	30 KM/Hr
30	Blind Spot	Blind Spot	11.77258, 78.05460	10 KM/Hr
31	Turn	High	11.77233, 78.05483	15 KM/Hr
32	Blind Spot	Blind Spot	11.76570, 78.05270	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
2	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
3	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
4	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
5	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
6	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
7	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
10	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
11	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
12	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
13	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
14	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
15	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
16	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
17	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
18	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
19	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
22	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
24	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
9	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
23	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46048, 77.85036



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45631, 77.81668



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49409, 77.88004



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49419, 77.88015



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49419, 77.88454



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49448, 77.88490



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.55608, 77.89762



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58320, 77.90646



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.64197, 77.95197



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.68921, 77.97737



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.68927, 77.97749



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.70223, 77.97860



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.71918, 78.01127



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72565, 78.01770



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73651, 78.03873



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73658, 78.03897



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74271, 78.04717



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.77258, 78.05460



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.77233, 78.05483



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.76570, 78.05270
